Hybrid Wind-Solar Energy Powered EV Charging Station

M³FluiD Lab
Multiscale Multiphysics Modeling and Data-Driven Analysis of Thermofluids Lab

Undergraduate Student Research Award (USRA) - 2023
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Abstract

This project explores the development of a hybrid charging station in Cleveland, Ohio, which utilizes both wind and solar energy to charge electric vehicles. The primary objective is to assess the feasibility of a charging station that harnesses the advantages of both renewable energy sources. To achieve this, we collect data on wind speed and solar irradiance in Cleveland to gauge the potential energy generation. Through calculations, we aim to ascertain the station's energy requirements for daily operation and its capacity to effectively charge electric cars within the specified timeframe.

Data Collection

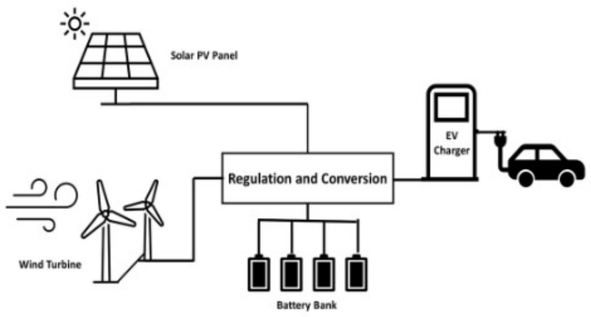
- Weather data is collected from www.visualcrossing.co
- Wind data collection every 10 minutes.
- Solar data collection every hour.
- Both wind and solar data collection over a period of 10 years, 2013 to 2022.
- The data was then analyzed and studied using MATLAB and Excel to create tables and graphs showing weather trends for every hour, day, month, and year.
- An optimum wind turbine and PV solar panel are selected to power a level-2 EV charging station.

Assumptions

- The Level-2 EV charging station model J1772 which has a required power of 11.5 kWh.
- The EV is the Tesla Model Y which has an 80-kW battery capacity and a range of 330 miles per full charge.

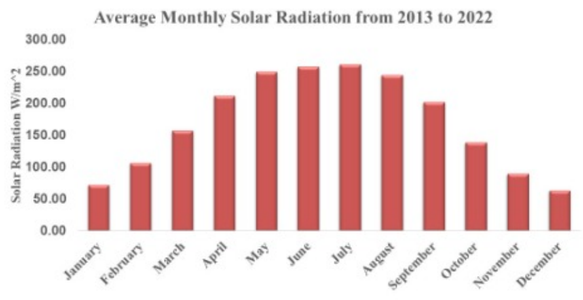
Methodology

- Analyze wind and solar data to determine the share of wind and solar available in Cleveland.
- Select a wind turbine and PV solar based on the average energy densities.
- Utilize a hybrid wins-solar system to provide the energy needed to power a Level-2 EV charging station with 24-hour availability.



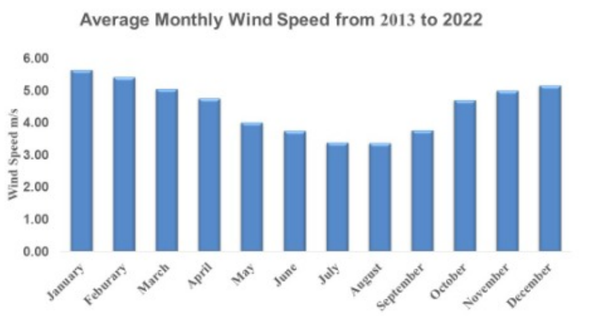
Results - Solar

- The maximum monthly mean solar radiations are in June, July, and August.
- The minimum monthly mean solar radiations are in January and December.
- Average solar irradiation in Cleveland, Ohio is 164.43 W/m²



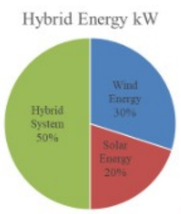
Results - Wind

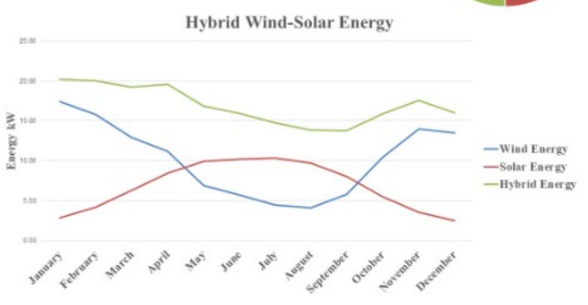
- The maximum monthly mean wind speeds are in January, February, and December.
- The minimum monthly mean wind speeds are in July and August.
- Average wind speed is 4.68 m/s



Results - Hybrid System

- The hybrid wind-solar system can generate an average energy output of 17 kWh.
- The hybrid system generates enough energy to power the level 2 EV charging stations.
- The system can fully charge 3 EVs during the day.





Discussion

- Feasibility and Advantages: The research confirms the feasibility of a hybrid wind-solar EV charging station, harnessing renewable energy, reducing emissions, and promoting sustainability.
- Challenges and Solutions: Intermittent weather challenges are addressed using advanced forecasting techniques.
- Future Research: Future studies should explore scalability, energy storage integration, and conduct economic analyses for enhanced performance and long-term financial benefits.

Conclusion

- This research demonstrates the feasibility of utilizing a hybrid wind-solar system to provide power for a level-2 EV Charging Station in Cleveland, Ohio.
- The Research represents a promising solution for sustainable EV charging infrastructure.
- Embracing such innovative solutions can accelerate the shift towards a greener and more sustainable transportation ecosystem.
- More analysis on the hybrid wind-solar systems is needed in the following phases of this research.

Acknowledgment

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